Algebra: Simultaneous Equations

Questions 1–14

Question 1. Solve the following sets of simultaneous equations:

a) $2x + 3y = 5$ and $7x - y = 6$
b) $6x = 5 - 2y$ and $x - 2y = 2$
c) $\frac{x-9}{3} + y = 0$ and $2(x + y) = 2$
d) $3x - y = 4$ and $\frac{y}{2} + \frac{x}{4} = 5$

Question 2. It is given that $y - kx = 10$ and $y = 2x - 10$, where k is an unknown constant, intersect when $y = 15$. Find

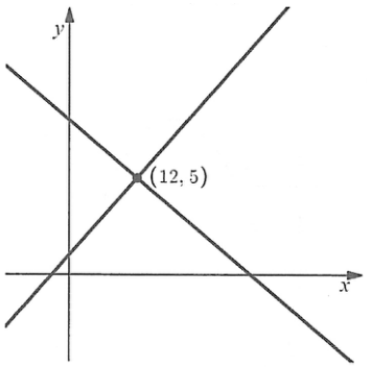
a) the x-coordinate of the point of intersection,
b) the value of k

Question 3. Two lines are given to be $y + ax = 5$ and $\frac{y}{b} + \frac{11x}{80} = 1$ where a and b are unknown constants. The two lines never intersect, and one line passes through the y-axis at $y = 8$.

a) the value of b

b) the value of a

Question 4. The diagram below shows two lines, which intersect at the point $(12, 5)$.



Given one line has gradient $-\frac{1}{4}$ and the other line intersects the y-axis at $y = 1$, find the equations of both lines in the form $ay + bx = c$ where a, b and c are integers.

Question 5. Solve the following sets of simultaneous equations:

a) $y = 2x + 4$ and $y = x^2 + 1$

b) $y = 3x - 2$ and $y = x^2 - 4x - 10$

c) $y = \frac{1}{2}x$ and $y = x^2 + 2xy$

Question 6. Two equations are given by $3y - x + 10 = 0$ and $x^2 + y^2 = 10$.

a) Show that the two equations may be satisfied by $x^2 - 2x + 1 = 0$.

b) Hence solve the simultaneous equations $3y - x + 10 = 0$ and $x^2 + y^2 = 10$.

Question 7. Solve the following sets of simultaneous equations:

a) $y + 3x = 15$ and $y^2 = 9(x - 3)$

b) $y - x = 4$ and $x^2 + y = 10$

c) $y - 2x = 12$ and $x^2 + 2xy = 5$

Question 8. Solve the following sets of simultaneous equations:

a) $y + x = 4$ and $x^2 + y^2 = 16$

b) $y + 4x = 6$ and $x^2 + 2x + y^2 - 3y = 1$

c) $2y + x = 8$ and $x^2 + y^2 - xy = 12$

d) $x = 2y + 1$ and $y^2 + 8 = 3xy$

Question 9. Two equations are given by $x^2 + 2x + y^2 = 24$ and $2y = x - 12$. Show that there are no solutions when solving simultaneously.

Question 10. The curve $y = px^2 - qx + 1$ where p and q are positive constants, passes through the points $(5, 36)$ and $(-2, 15)$. Find the values of p and q.

Question 11. The curve $y = px^3 - 5x^2 + x + q$ where p and q are positive constants, passes through the points $(1, 6)$ and $(-\frac{1}{2}, \frac{21}{4})$. Find the values of p and q.

Question 12. Find the two positive real numbers with a product of 20 and a difference of $\frac{8}{3}$.

Question 13. Solve the simultaneous equations $2^{x-1} = 4^{2y}$ and $9^{x-2} = 3^{y+1}$.

Question 14. Solve the simultaneous equations $5^{2x-3} = 25^{y-1}$ and $16^{x-1} = 2^{3y}$.